

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

02nd December 2020

Subject SP8 – General Insurance: Pricing

Time allowed: 3 Hours 30 Minutes (14.30 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
- 3. Mark allocations are shown in brackets.*

- Q. 1)** You are the pricing head of “MotoSure”, an insurance company, which specializes in providing Motor insurance. The Company offers an add-on to its Fleet portfolio, which protects the loss of income on the days when the fleet does not operate.

The Company has been offering the add-on for past 6 years and an individual fleet is usually priced considering its loss experience of past 3 years.

- i) Recently the country in which the MotoSure operates announced lock-down due to which many of the fleets could not ply for 2 months, thereby resulting in higher claims for the insurer under the add-on. There is a fleet which has been insuring with the insurer for the past 2 years and is due for renewal post lockdown. Please discuss the method to be opted by the insurer to price the add-on for this fleet. (5)
- ii) It is observed that for a particular category of vehicles, motor OD claims follow Compound binomial distribution with expected claim number of 1000 per year, probability of claim of 0.05. The claim amounts follow Log Normal distribution with $\mu = 8.2198$, $\sigma = 1.4075$. Calculate standard for full credibility such that the observed aggregate loss is within a proportion of $\pm 5\%$ of the mean with 90% probability. (4)
- iii) It has always been a practice for insurer to cap the losses / remove the large losses from data while pricing. From credibility theory point of view, describe how does this help to improve the credibility of data. (3)

[12]

- Q. 2)** Outline the differences between pricing a risk by the insurer and Reinsurer. [5]

- Q. 3)** A mid-size general insurer wants to expand its portfolio of burglary insurance. The insurer has both policy and claims data for 6 years before those of past 2 years. You have been asked to use Frequency – Severity approach to price the same.

- i) Describe the adjustments needed to the base data available to price the policy. (8)
- ii) One of your team members is suggesting to use Burning cost approach instead of Frequency – Severity approach. Explain why Frequency Severity may be more / less advantageous than Burning cost approach. (4)
- iii) Briefly discuss the issues to consider after arriving at the technical premium for the property damage cover of a Commercial property. (3)

[15]

- Q. 4)** You are working as pricing head for an insurer, which writes motor insurance and uses GLM model to determine risk premium for motor portfolio.

- i) Describe Aliasing and explain with suitable example. (2)
- ii) The Motor underwriter has suggested adding the 3-year average annual mileage of the vehicle as a rating factor to differentiate risks. Discuss the effectiveness of the rating factor proposed by the underwriter. (3)
- iii) Recently the cyclonic activities have increased due to global warming. You have studied the impact of the recent cyclones on the risk premium and need to allow for the same across various pincode locations across the country. Explain which method will suit this exercise. (2)

- iv) You have fitted 2 models to arrive at the severity from the claims data available. Following are the model predictions for vehicle with IDV Rs 1,00,000 :

No. of claims	Claim amount (₹)	
	Model 1	Model 2
3	40000	24000
1	2000	1500
2	3000	2200
1	2667	3000
1	1428	1600
2	1500	4000
2	700	5500
3	12000	14000
1	22000	40000
1	15000	16000

Explain how to validate which model is more effective at distinguishing high and low experience. Also, enlist the steps involved in the validation.

(3)
[10]

- Q. 5)** i) List the considerations for selecting appropriate ILF table for casualty line of business. (4)

- ii) Explain how the ILF curve for casualty line of business varies from those of Property line of business. (2)

- iii) Which set of curves are mostly used for Casualty line of business and briefly mention the reasons for the same. (2)
[8]

- Q. 6)** A country has three categories of motor vehicles i.e. Private Car, Commercial vehicle and two-wheeler. Large number of 2-wheeler vehicles are uninsured despite low premiums for motor third party cover. The regulator has mandated all insurance companies must write minimum obligation of Motor Third party business in a given year which is based on total premium written by each company and by whole industry in this segment in past financial year. The Government and all insurance companies want to achieve the objective of 100% insurance in this motor third party segment for all categories of vehicle over the period.

- i) Outline the concerns various stakeholders may have in the current method of calculating the minimum obligation which is impediment to achieve its objective. (4)

- ii) Describe the steps Government can take to achieve its objectives and suggest alternative method of calculating minimum obligation. (6)
[10]

- Q. 7)** Discuss the factors that the Government should consider in establishing a central data repository in respect of Motor portfolio to support the data needs of the Industry. [10]

- Q. 8)** i) Discuss the possible actuarial investigations in determining appropriateness of reinsurance arrangements and the various aspects of the reinsurance requirements of the company. (6)

- ii)** Describe source of uncertainty in a model specification. (4)
[10]
- Q. 9) i)** Define risk factors and rating factors. (2)
- ii)** An Insurance company is seeking reinsurance cover for it's medical malpractice business and broker has requested additional details around number of doctors covered, specialty from the insurance company beyond premium volumes to price their reinsurance cover. Explain the need for additional details requested by the broker. (3)
[5]
- Q. 10) i)** List the perils covered under Cyber insurance product. (5)
- ii)** An Insurance company is exploring purchase of a cyber-catastrophe model from an external model provider to enable them price aggregate exposure to cyber risk. Discuss the challenges with pricing based upon current cyber catastrophe models. (5)
- iii)** List possible underwriting factors for public liability insurance cover. (5)
[15]
